WORKED SOLUTION

STEP I 2012, Question 6

Pure • Trigonometric Inequalities

STEP I 2012, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q6

The key idea

Every angle of elevation here has the same opposite side, the flagpole, so each cotangent is just a horizontal distance divided by the height. The first result then becomes a fact about lengths in the circle: $DC^2 = AD \cdot DB$, because the angle in a semicircle is a right angle. The trigonometric identity turns $\cot p \cot q - \cot^2 \frac{1}{2}(p+q)$ into something whose sign we can read off. Once we have that inequality, the geometry hands us the final result almost for free.

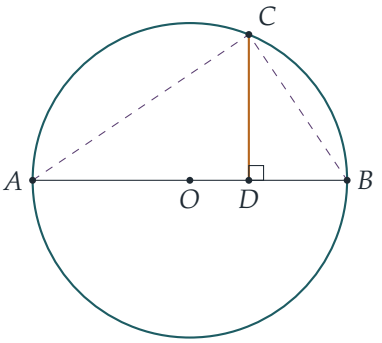
Solution

The geometric result

Let the flagpole have height $h > 0$ and top T, so DT is vertical. Each of A, B, C is on the ground, so the triangles TDA, TDB, TDC have right angles at D. Hence

$$\cot\alpha = \frac{AD}{h}, \qquad \cot\beta = \frac{DB}{h}, \qquad \cot\phi = \frac{DC}{h}.$$

Since α and β are acute, D is neither A nor B (from either of those points the elevation would be a right angle), so D lies strictly between A and B and $DC > 0$.



Now look at the ground plan. Let O be the centre, r the radius and $x = OD$ (measured towards B, so $-r < x < r$). Then $AD = r + x$ and $DB = r - x$, and by Pythagoras in triangle ODC, $DC^2 = OC^2 - OD^2 = r^2 - x^2$. Therefore

$$AD \cdot DB = (r + x)(r - x) = r^2 - x^2 = DC^2.$$

(Equivalently: $\angle ACB$ is a right angle, being the angle in a semicircle, and CD is the altitude of the right-angled triangle ACB.) Dividing by h^2,

$$\cot\alpha \cot\beta = \frac{AD \cdot DB}{h^2} = \frac{DC^2}{h^2} = \cot^2\phi,$$

as required.

The identity

Write $s = \frac{1}{2}(p + q)$ and use the double-angle forms $\sin^2 s = \frac{1}{2}(1 - \cos(p + q))$ and $\cos^2 s = \frac{1}{2}(1 + \cos(p + q))$. The left-hand side becomes

$$\begin{aligned}
&\tfrac{1}{2}\cos p \cos q(1 - \cos(p + q)) - \tfrac{1}{2}\sin p \sin q(1 + \cos(p + q)) \\
&\quad = \tfrac{1}{2}(\cos p \cos q - \sin p \sin q) - \tfrac{1}{2}\cos(p + q)(\cos p \cos q + \sin p \sin q) \\
&\quad = \tfrac{1}{2}\cos(p + q) - \tfrac{1}{2}\cos(p + q)\cos(p - q),
\end{aligned}$$

using the compound-angle formulae for $\cos(p + q)$ and $\cos(p - q)$. This holds for all p and q.

The inequality

Factorise the right-hand side of the identity:

$$\cos p \cos q \sin^2 s - \sin p \sin q \cos^2 s = \tfrac{1}{2}\cos(p + q)(1 - \cos(p - q)).$$

Now suppose $p, q > 0$ and $p + q \leq \frac{1}{2}\pi$. Then $0 < p + q \leq \frac{1}{2}\pi$, so $\cos(p + q) \geq 0$; and $1 - \cos(p - q) \geq 0$ always. So the right-hand side is non-negative, and

$$\cos p \cos q \sin^2 s \;\geq\; \sin p \sin q \cos^2 s.$$

Ask yourself Can we safely divide through by $\sin p \sin q \sin^2 s$?

Yes: since p and q are positive with sum at most $\frac{1}{2}\pi$, each of p, q and s lies in $(0, \frac{1}{2}\pi)$, so $\sin p$, $\sin q$ and $\sin s$ are all strictly positive. Dividing by the positive quantity $\sin p \sin q \sin^2 s$ keeps the direction of the inequality:

$$\cot p \cot q \;\geq\; \cot^2 s = \cot^2 \tfrac{1}{2}(p + q).$$

Equality holds exactly when $p = q$ or $p + q = \frac{1}{2}\pi$.

The final result

Take $p = \alpha$, $q = \beta$. Both are positive (they are angles of elevation of a pole of positive height) and $\alpha + \beta \leq \frac{1}{2}\pi$, so the inequality applies. Combined with the first result,

$$\cot^2 \phi = \cot\alpha \cot\beta \;\geq\; \cot^2 \tfrac{1}{2}(\alpha + \beta).$$

Both ϕ and $\frac{1}{2}(\alpha + \beta)$ lie in $(0, \frac{1}{2}\pi)$ (ϕ because $DC > 0$ and $h > 0$), so both cotangents are positive and we may take square roots: $\cot\phi \geq \cot\frac{1}{2}(\alpha + \beta)$.

? **Ask yourself** Why does this give $\phi \leq \frac{1}{2}(\alpha + \beta)$ and not $\geq$?

Because cot is strictly *decreasing* on $(0, \frac{1}{2}\pi)$: a larger cotangent means a smaller angle. Hence

$$\phi \leq \tfrac{1}{2}(\alpha + \beta) \qquad \text{whenever } \alpha + \beta \leq \tfrac{1}{2}\pi.$$

Checks

- If $D = O$ then $AD = DB = DC = r$, so $\alpha = \beta = \phi$: both sides of the first result are equal, and the final inequality is an equality, matching the case $p = q$.
- A numerical test: $r = 1$, $OD = \frac{1}{2}$, $h = 1$ gives $\cot\alpha = 1.5$, $\cot\beta = 0.5$, $\cot^2\phi = 0.75$, as predicted. Here $\alpha + \beta \approx 0.588 + 1.107 = 1.695 > \frac{1}{2}\pi$, so the final result need not apply; and indeed $\phi \approx 0.857 > \frac{1}{2}(\alpha + \beta) \approx 0.848$, showing the condition matters.
- We checked the identity at 10^5 random (p, q), the inequality at 2×10^5 random pairs in the allowed range, and $\phi \leq \frac{1}{2}(\alpha + \beta)$ in about 8×10^4 random configurations with $\alpha + \beta \leq \frac{1}{2}\pi$, with no failures.

Common slips

- Forgetting that cot is decreasing, and so concluding $\phi \geq \frac{1}{2}(\alpha + \beta)$.
- Dividing the inequality by $\sin p \sin q \sin^2 s$ without saying why it is positive.
- Ignoring the condition $p + q \leq \frac{1}{2}\pi$: it is exactly what makes $\cos(p + q) \geq 0$, and without it the inequality can fail.

Beyond STEP

The inequality says that $\ln\cot t$ is a convex function on $(0, \frac{1}{4}\pi]$, since it states that the average of $\ln\cot p$ and $\ln\cot q$ is at least $\ln\cot$ of the average. Convexity and Jensen's inequality are a powerful source of inequalities in first-year analysis. The relation $DC^2 = AD \cdot DB$ is Euclid's construction of a geometric mean: $\cot\phi$ is the geometric mean of $\cot\alpha$ and $\cot\beta$. Bartle and Sherbert's *Introduction to Real Analysis* treats convex

functions, and Siklos's *Advanced Problems in Mathematics* has more questions in this style.

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